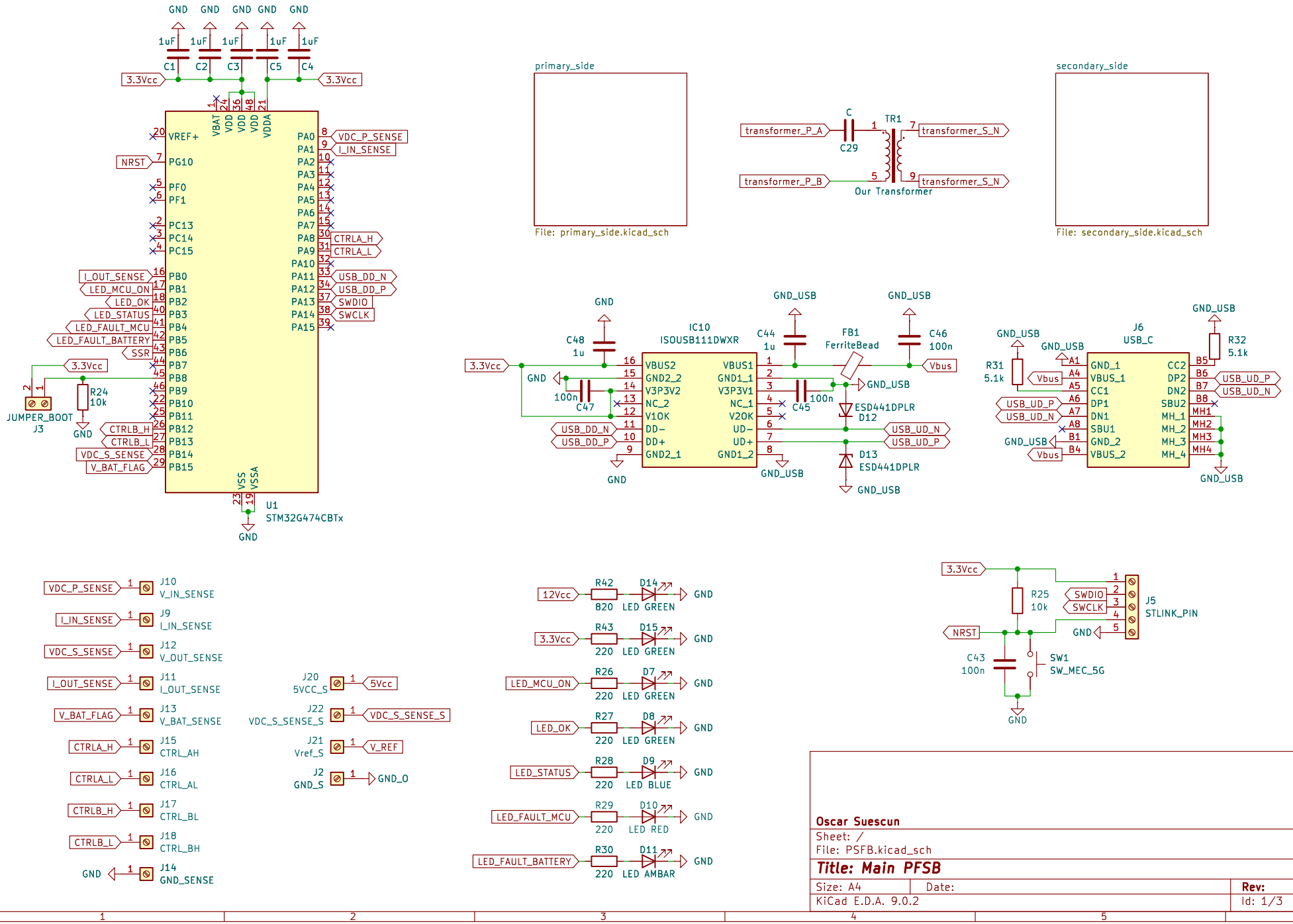
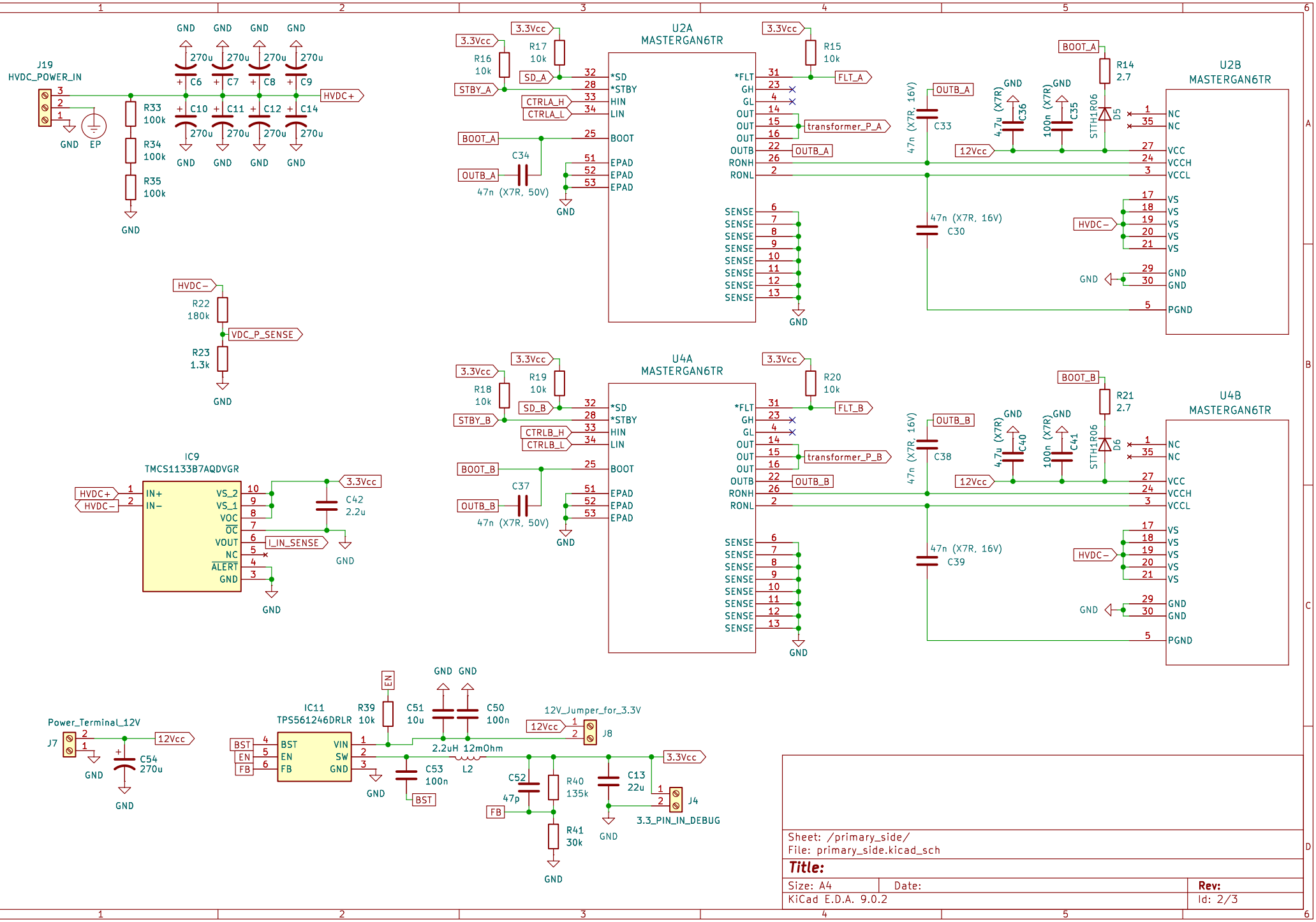
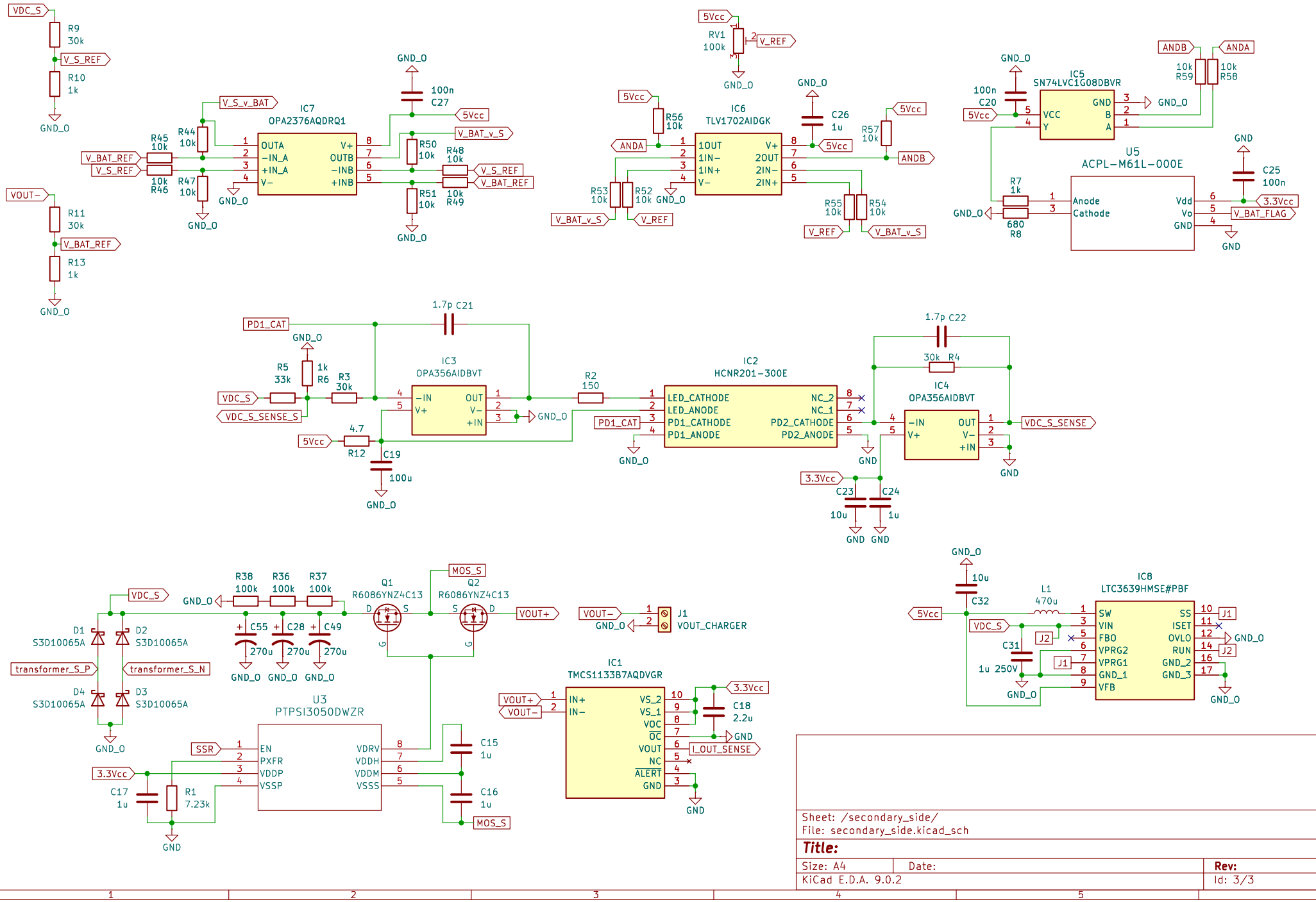

GND GND GND GND GND
1uF 1uF 1uF 1uF 1uF
C1 C2 C3 C5 C4
3.3Vcc 3.3Vcc
VBAT VDD VDD VDD VDDA
VREF+ NRST PG10 PF0 PF1 PC13 PC14 PC15 PB0 PB1 PB2 PB3 PB4 PB5 PB6 PB7 PB8 PB9 PB10 PB11 PB12 PB13 PB14 PB15
PA0 PA1 PA2 PA3 PA4 PA5 PA6 PA7 PA8 PA9 PA10 PA11 PA12 PA13 PA14 PA15
VDC_P_SENSE I_IN_SENSE CTRLA_H CTRLA_L USB_DD_N USB_DD_P SWDIO SWCLK
I_OUT_SENSE LED_MCU_ON LED_OK LED_STATUS LED_FAULT_MCU LED_FAULT_BATTERY SSR
3.3Vcc
JUMPER_BOOT
J3
R24
10k
GND
CTRLB_H CTRLB_L VDC_S_SENSE V_BAT_FLAG
VSS VSSA
U1
STM32G474CBTx
GND
primary_side
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transformer_P_A
transformer_P_B
C
C29
TR1
transformer_S_N
transformer_S_N
Our Transformer
secondary_side
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GND
C48
1u
IC10
ISOUSB111DWXR
VBUS2 GND2_2 V3P3V2 NC_2 V1OK DD- DD+ GND2_1
VBUS1 GND1_1 V3P3V1 NC_1 V2OK UD- UD+ GND1_2
3.3Vcc
GND
100n
C47
USB_DD_N
USB_DD_P
GND
GND_USB
C44
1u
FB1
FerriteBead
C46
100n
Vbus
GND_USB
100n
C45
ESD441DPLR
D12
USB_UD_N
USB_UD_P
D13
ESD441DPLR
GND_USB
GND_USB
J6
USB_C
GND_1 VBUS_1 CC1 DP1 DN1 SBU1 GND_2 VBUS_2
CC2 DP2 DN2 SBU2 MH_1 MH_2 MH_3 MH_4
A1 A4 A5 A6 A7 A8 B1 B4
B5 B6 B7 B8 MH1 MH2 MH3 MH4
R31
5.1k
R32
5.1k
Vbus
USB_UD_P
USB_UD_N
GND_USB
Vbus
GND_USB
VDC_P_SENSE
J10
V_IN_SENSE
I_IN_SENSE
J9
I_IN_SENSE
VDC_S_SENSE
J12
V_OUT_SENSE
I_OUT_SENSE
J11
I_OUT_SENSE
V_BAT_FLAG
J13
V_BAT_SENSE
CTRLA_H
J15
CTRL_AH
CTRLA_L
J16
CTRL_AL
CTRLB_H
J17
CTRL_BL
CTRLB_L
J18
CTRL_BH
GND
J14
GND_SENSE
J20
5VCC_S
5Vcc
J22
VDC_S_SENSE_S
VDC_S_SENSE_S
J21
Vref_S
V_REF
J2
GND_S
GND_O
12Vcc
R42
820
D14
LED GREEN
GND
3.3Vcc
R43
220
D15
LED GREEN
GND
LED_MCU_ON
R26
220
D7
LED GREEN
GND
LED_OK
R27
220
D8
LED GREEN
GND
LED_STATUS
R28
220
D9
LED BLUE
GND
LED_FAULT_MCU
R29
220
D10
LED RED
GND
LED_FAULT_BATTERY
R30
220
D11
LED AMBAR
GND
3.3Vcc
R25
10k
NRST
C43
100n
GND
SWDIO
SWCLK
GND
J5
STLINK_PIN
SW1
SW_MEC_5G
GND
Oscar Suescun
Sheet: /
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Title: Main PFSB
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J19
HVDC_POWER_IN
GND
GND
GND
GND
270u
C6
C7
C8
C9
C10
C11
C12
C14
HVDC+
R33
100k
R34
100k
R35
100k
EP
HVDC−
R22
180k
VDC_P_SENSE
R23
1.3k
U2A
MASTERGAN6TR
3.3Vcc
R16
10k
R17
10k
R15
10k
SD_A
STBY_A
CTRLA_H
CTRLA_L
BOOT_A
OUTB_A
C34
47n (X7R, 50V)
*SD
*STBY
HIN
LIN
BOOT
EPAD
*FLT
GH
GL
OUT
OUTB
RONH
RONL
SENSE
FLT_A
transformer_P_A
OUTB_A
U2B
MASTERGAN6TR
BOOT_A
R14
2.7
STTH1R06
D5
C33
47n (X7R, 16V)
C36
4.7u (X7R)
C35
100n (X7R)
12Vcc
C30
47n (X7R, 16V)
NC
VCC
VCCH
VCCL
VS
GND
PGND
IC9
TMCS1133B7AQDVGR
HVDC+
HVDC−
IN+
IN−
VS_2
VS_1
VOC
OC
VOUT
NC
ALERT
GND
C42
2.2u
I_IN_SENSE
U4A
MASTERGAN6TR
R18
10k
R19
10k
R20
10k
SD_B
STBY_B
CTRLB_H
CTRLB_L
BOOT_B
OUTB_B
C37
47n (X7R, 50V)
transformer_P_B
OUTB_B
U4B
MASTERGAN6TR
BOOT_B
R21
2.7
D6
C38
C40
C41
C39
Power_Terminal_12V
J7
C54
270u
IC11
TPS561246DRLR
R39
10k
EN
C51
10u
C50
100n
12V_Jumper_for_3.3V
J8
BST
VIN
SW
2.2uH 12mOhm
L2
C53
100n
C52
47p
R40
135k
R41
30k
C13
22u
J4
3.3_PIN_IN_DEBUG
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VDC_S
R9
30k
V_S_REF
R10
1k
GND_O
VOUT−
R11
30k
V_BAT_REF
R13
1k
GND_O
V_S_v_BAT
IC7
OPA2376AQDRQ1
R44
10k
R45
10k
V_BAT_REF
V_S_REF
10k
R46
R47
10k
GND_O
OUTA
−IN_A
+IN_A
V−
V+
OUTB
−INB
+INB
GND_O
100n
C27
5Vcc
V_BAT_v_S
R50
10k
R48
10k
V_S_REF
V_BAT_REF
10k
R49
R51
10k
GND_O
5Vcc
RV1
100k
V_REF
GND_O
5Vcc
R56
10k
ANDA
R53
10k
R52
10k
GND_O
V_BAT_v_S
V_REF
IC6
TLV1702AIDGK
1OUT
1IN−
1IN+
V−
V+
2OUT
2IN−
2IN+
GND_O
C26
1u
5Vcc
R57
10k
ANDB
R55
10k
R54
10k
V_REF
V_BAT_v_S
GND_O
100n
C20
5Vcc
IC5
SN74LVC1G08DBVR
VCC
Y
GND
B
A
GND_O
ANDB
ANDA
10k
R59
10k
R58
U5
ACPL−M61L−000E
R7
1k
GND_O
680
R8
Anode
Cathode
Vdd
Vo
GND
GND
C25
100n
3.3Vcc
V_BAT_FLAG
GND
PD1_CAT
GND_O
1.7p C21
R5
33k
1k
R6
R3
30k
VDC_S
VDC_S_SENSE_S
IC3
OPA356AIDBVT
−IN
V+
OUT
V−
+IN
GND_O
4.7
5Vcc
R12
C19
100u
GND_O
R2
150
IC2
HCNR201−300E
LED_CATHODE
LED_ANODE
PD1_CATHODE
PD1_ANODE
NC_2
NC_1
PD2_CATHODE
PD2_ANODE
PD1_CAT
GND_O
1.7p C22
30k R4
IC4
OPA356AIDBVT
−IN
V+
OUT
V−
+IN
VDC_S_SENSE
GND
3.3Vcc
C23
10u
C24
1u
GND
GND
R38
100k
R36
100k
R37
100k
GND_O
VDC_S
Q1
R6086YNZ4C13
Q2
R6086YNZ4C13
MOS_S
VOUT+
D1
S3D10065A
D2
S3D10065A
D4
S3D10065A
D3
S3D10065A
transformer_S_P
transformer_S_N
GND_O
C55
270u
C28
270u
C49
270u
GND_O
GND_O
GND_O
VOUT−
GND_O
J1
VOUT_CHARGER
U3
PTPSI3050DWZR
SSR
EN
PXFR
VDDP
VSSP
VDRV
VDDH
VDDM
VSSS
3.3Vcc
C17
1u
R1
7.23k
GND
C15
1u
C16
1u
MOS_S
IC1
TMCS1133B7AQDVGR
VOUT+
VOUT−
IN+
IN−
VS_2
VS_1
VOC
OC
VOUT
NC
ALERT
GND
3.3Vcc
C18
2.2u
GND
I_OUT_SENSE
GND
GND_O
10u
C32
5Vcc
L1
470u
IC8
LTC3639HMSE#PBF
VDC_S
C31
1u 250V
GND_O
J2
J1
SW
VIN
FBO
VPRG2
VPRG1
GND_1
VFB
SS
ISET
OVLO
RUN
GND_2
GND_3
J1
J2
GND_O
GND_O
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